Survey Integration Protocol (3)

PHASE 1 — Feedback Infrastructure Foundation

A. Objective

Create the foundational infrastructure for lightweight implicit feedback collection.

This phase establishes:

- event standards
- feedback schemas
- Firestore collections
- tracking helpers
- contextual UI framework
- summary aggregation pipelines

This matters because:

- all future implicit UX depends on standardized ingestion
- analytics and emotional signals must share a unified architecture
- reports become impossible if implicit data is inconsistent

B. Events Implemented in this Phase

B2 — Community Engagement

Event

```
implicit_reaction_selected
```

```
implicit_sentiment_tapped
```

B3 — Course Engagement

Event

```
content_helpful_tapped
```

```
content_not_helpful_tapped
```

```
lesson_confusion_signal_selected
```

```
lesson_continue_later_clicked
```

B5 — Friction / Drop-off

Event

```
implicit_feedback_dismissed
```

```
implicit_feedback_shown
```

B7 — Customer Profiles / Behavioral Understanding

Event

```
emotion_slider_changed
```

```
quick_sentiment_selected
```

C. Implementation TODOs

- create implicit feedback tracking helper
- create `survey_templates` collection
- create `survey_triggers` collection
- create `survey_responses` collection
- create `survey_summaries` collection
- create `trackImplicitFeedback()` wrapper
- implement contextual feedback UI component
- implement reaction chip component
- implement emotional slider component
- implement quick sentiment tap component
- implement Firestore write batching
- create Zod schema for implicit feedback payloads
- implement event registry additions
- create cooldown helper logic
- implement server timestamp normalization

- build aggregation trigger on `survey_responses`
- create daily bucket rollup aggregation
- create bucket score summary callable
- create dashboard query abstraction

D. How to Test in Emulator

Simulated User Flows

Flow 1 — Helpful content

1. Open lesson
2. Navigate several slides
3. Tap:

    > “Helpful”

4. Leave lesson

Expected System Behavior

- `content_helpful_tapped` written
- contextual metadata attached:
    - lesson_id
    - slide_index
    - session_id
    - course_id

Validation

Check:

```
analytics_events
survey_responses
survey_summaries
```

Verify:

- event timestamps
- user linkage
- bucket assignment
- deduplication

Flow 2 — Confusion signal

1. Fail quiz
2. Prompt appears:

   > “Still confused?”

3. User taps:

   > “Examples unclear”

Expected Behavior

- trigger logic executed
- survey instance generated
- contextual payload saved

E. Success Metrics

Metric	Goal
Feedback ingestion success	>99%
Duplicate response rate	<1%
Event normalization consistency	100%

Metric	Goal
Average UI latency	<100ms
Dashboard aggregation delay	<5 min
Crash increase from instrumentation	0

PHASE 2 — Behavioral Trigger Engine

A. Objective

Build intelligent contextual triggers so feedback only appears during meaningful behavioral moments.

This matters because:

- implicit feedback should feel assistive
- surveys should appear only when contextually justified
- reduces survey fatigue dramatically

B. Events Implemented in this Phase

B5 — Friction / Abandonment

Event

```
lesson_abandonment_feedback_triggered
```

```
quiz_confusion_feedback_triggered
```

```
navigation_dead_end_feedback_triggered
```

```
inactive_user_feedback_triggered
```

B6 — Conversion Timing

Event

```
checkout_hesitation_feedback_triggered
```

```
purchase_dropoff_feedback_triggered
```

B2 — Community Engagement

Event

```
social_hesitation_feedback_triggered
```

```
group_nonjoin_feedback_triggered
```

C. Implementation TODOs

- create trigger evaluation engine
- implement session-level event counting
- implement cooldown enforcement
- implement contextual trigger middleware

- implement inactivity detection
- implement repeated navigation loop detection
- implement lesson abandonment detection
- implement multi-failure quiz detection
- create user trigger history collection
- build trigger suppression logic
- implement trigger prioritization scoring
- create adaptive trigger selection logic
- implement experiment flag support

D. How to Test in Emulator

Flow 1 — Lesson abandonment

1. Open lesson
2. Advance several slides
3. Exit lesson before completion

Expected

- abandonment trigger evaluates
- feedback prompt shown once
- no repeat within cooldown window

Flow 2 — Navigation confusion

1. Bounce:

```
dashboard → curriculum → dashboard → curriculum
```

1. Repeat rapidly

Expected

- dead-end navigation trigger fires
- “Did you know where to go next?” appears

E. Success Metrics

Metric	Goal
Trigger precision	>80%
Trigger duplication rate	<5%
Survey fatigue rate	<10% dismiss immediately
Average prompts/session	<1
Cooldown compliance	100%

PHASE 3 — Cross-Platform Movement Intelligence

A. Objective

Measure whether users can successfully move through the Mortar ecosystem:

- curriculum
- community
- groups
- shop
- events
- networking

This matters because your platform value depends on ecosystem immersion, not isolated screen usage.

B. Events Implemented in this Phase

B3 — Curriculum Engagement

Event

```
curriculum_to_community_clicked
```

```
curriculum_to_event_clicked
```

```
curriculum_to_shop_clicked
```

B2 — Community Engagement

Event

```
community_to_profile_clicked
```

```
community_to_group_clicked
```

```
community_to_event_clicked
```

B6 — Funnel / Conversion

Event

```
cross_platform_transition_completed
```

```
cross_platform_transition_failed
```

B7 — Behavioral Mapping

Event

```
ecosystem_depth_score_updated
```

```
platform_meander_detected
```

C. Implementation TODOs

- create cross-platform navigation tracker
- implement ecosystem session graph builder
- implement platform transition event mapping
- create ecosystem depth scoring logic
- implement dead-end route detection
- create transition success aggregation
- build movement heatmap summaries
- create “next action clarity” implicit feedback prompt

- implement navigation path persistence
- create user ecosystem engagement summary

D. How to Test in Emulator

Flow 1 — Healthy ecosystem movement

1. Dashboard
2. Curriculum
3. Community
4. Event
5. Group

Expected

- transition graph generated
- ecosystem depth score increases
- no confusion prompts triggered

Flow 2 — Dead-end movement

1. Dashboard
2. Curriculum
3. Exit app

Repeated 4 sessions.

Expected

- low ecosystem integration score
- pathway weakness surfaced

E. Success Metrics

Metric	Goal
Cross-platform tracking completeness	>95%
Ecosystem path reconstruction accuracy	>90%
Dead-end route identification usefulness	actionable
Transition latency impact	negligible

PHASE 4 — Implicit Emotional Intelligence Layer

A. Objective

Measure emotional UX states without explicit surveys.

This phase answers:

- overwhelm
- confidence
- clarity
- motivation
- belonging

without asking long-form questions.

B. Events Implemented in this Phase

B5 — Friction

Event

```
overwhelmed_sentiment_selected
```

```
confused_sentiment_selected
```

B4 — Skill Growth

Event

```
confidence_growth_signal_selected
```

```
skill_clarity_signal_selected
```

B2 — Community

Event

```
community_belonging_signal_selected
```

```
networking_confidence_signal_selected
```

C. Implementation TODOs

- create emotional sentiment taxonomy
- implement emoji-based sentiment chips

- create confidence progression summaries
- implement emotional trend aggregation
- create overwhelm scoring engine
- implement emotional-state timeline summaries
- build emotional heatmap dashboard
- create AI interpretation layer
- implement cohort emotional comparisons
- create longitudinal emotional analytics

D. How to Test in Emulator

Flow 1 — Overwhelmed learner

1. Fail quiz repeatedly
2. Long session duration
3. Tap:

    > “Overwhelmed”

Expected

- emotional trend updated
- frustration correlation stored
- cohort metrics updated

E. Success Metrics

Metric	Goal
Emotional feedback participation	>35%
Sentiment-to-behavior correlation usefulness	high

Metric	Goal
Emotional trend consistency	stable
User complaint rate about prompts	very low

PHASE 5 — Admin Intelligence Reports

A. Objective

Transform implicit feedback + analytics into actionable operational intelligence.

This phase answers:

- Why users disengage
- Why users convert
- Why users fail to integrate socially
- Why users stop progressing

B. Events Implemented in this Phase

Derived / Backend

Event

```
derived_social_engagement_score
```

```
derived_conversion_friction_score
```

```
derived_ecosystem_integration_score
```

```
derived_emotional_health_score
```

C. Implementation TODOs

- create executive analytics dashboard
- build bucket scoring engine
- create friction explanation reports
- implement emotional trend reports
- create conversion explanation reports
- create ecosystem movement heatmaps
- implement cohort comparison dashboards
- create AI insight summarization
- implement anomaly detection
- build recommendation engine

D. How to Test in Emulator

Flow 1 — Simulated struggling cohort

Simulate:

- high abandonment
- low purchases
- low social participation
- high confusion signals

Expected Dashboard Results

Admin should clearly see:

- primary friction points
- emotional degradation
- broken pathways
- weak conversion zones

E. Success Metrics

Metric	Goal
Report usefulness	actionable
Admin interpretation time	low
Root-cause identification accuracy	high
Dashboard load performance	<3s

FINAL REPORTS THE SYSTEM SHOULD ANSWER

1. Social Engagement Intelligence

Questions answered:

- Why don’t users interact?
- Which groups create retention?
- Which users become socially integrated?
- What causes community hesitation?

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2. Conversion Intelligence

Questions answered:

- Why do users abandon purchases?
- What content increases purchase likelihood?

- Which emotional states correlate with purchases?
- Which ecosystem pathways produce revenue?

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3. Ecosystem Integration Intelligence

Questions answered:

- Can users navigate naturally?
- Which routes produce retention?
- Which areas become dead ends?
- Are users immersing themselves in the ecosystem?

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4. Emotional UX Intelligence

Questions answered:

- Where do users feel overwhelmed?
- Which lessons reduce confidence?
- Which experiences create motivation?
- How emotionally healthy is the platform?